



Since June 2020, the Food and Agriculture Organization of the United Nations' (FAO's) **DIEM** has informed decision-making with regularly collected and analysed data in support of agricultural livelihoods. With its four pillars, DIEM explores the impact of shocks in the context of food crises, and brings together information and analysis in the DIEM Hub through a series of products and tools.

In order to inform decision-making, DIEM provides the answers to the following questions:

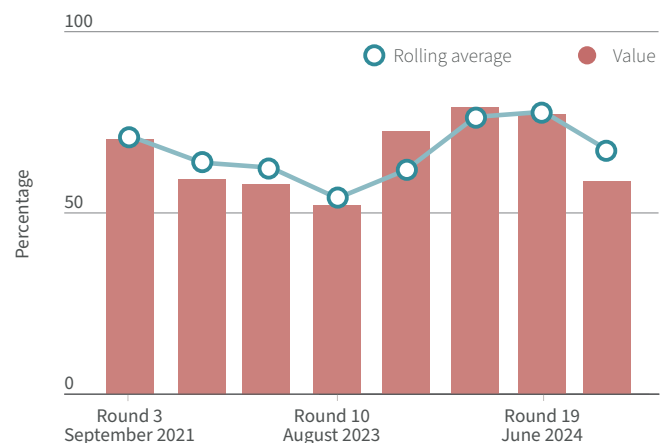
- What has been and is most likely to be the impact of shocks on agriculture?
- Where are the agricultural households most in need?
- Who are the agricultural households most in need?
- When are the needs of agricultural households the greatest?
- What would be the potential impact if the needs of agricultural households are not addressed?
- What is the value of damages, losses and recovery needs for the agriculture sector?

DIEM-Monitoring

In 29 of the world's most food-insecure countries, DIEM-Monitoring enumerators collect data at household level on shocks, agricultural livelihoods, food security and needs several times a year through computer-assisted telephone interviews and face-to-face surveys. This regularly collected and granular data is easily accessible in the form of dashboards, maps, briefs and aggregated data on the Hub, enabling partners

and stakeholders to trigger immediate mitigation and response actions. Following years of consistent data collection and analysis across time and countries, DIEM has developed the **Trend Analysis Application** to assist users in quickly identifying temporal trends for each indicator (an example of this is provided in Figure 1). Agricultural input trader surveys have also started in some countries and contribute to monitoring local value chains in addition to household surveys.

Figure 1. Crop harvest decrease trend over time in Yemen (2021–2024)



Source: FAO. 2024. Data in Emergencies: Yemen assessments results (2021–2024). [Accessed on 19 December 2024]. <https://data-in-emergencies.fao.org/apps/ff8806ac2ee14e1b9da3a8297edf7f37/explore> Licence: CC-BY-4.0.

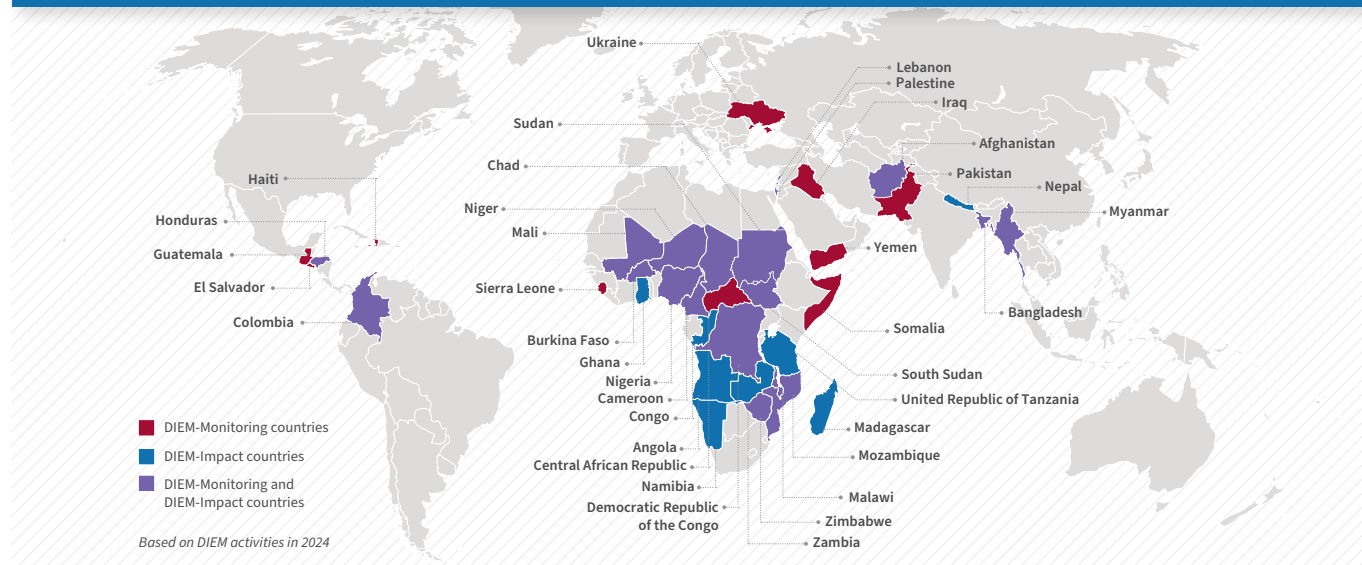
DIEM-Impact

DIEM-Impact takes the form of *ex post* assessments following large-scale hazards such as floods, drought or conflict. Combining remote sensing data, household-level data, secondary data and qualitative information, DIEM identifies agricultural populations and areas affected, determines the value of damages and losses in the agriculture sector, and lays the foundations of emergency and recovery activities.



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Figure 2. Countries with DIEM-Monitoring and DIEM-Impact activities in 2024



Note: Refer to the disclaimer on this page for the names and boundaries used in this map. Final boundary between the Republic of the Sudan and the Republic of South Sudan has not yet been determined. Final status of the Abyei area is not yet determined. Dotted line represents approximately the Line of Control in Jammu and Kashmir agreed upon by India and Pakistan. The final status of Jammu and Kashmir has not yet been agreed upon by the parties. **Source of data:** FAO. 2024. DIEM-Monitoring and DIEM-Impact. In: *Data in Emergencies (DIEM) Hub*. Rome. [Cited 11 December 2024]. <https://data-in-emergencies.fao.org>
Source of map: United Nations Geospatial. 2025. Map of the World. In: *United Nations*. [Cited 3 April 2025]. <https://www.un.org/geospatial/content/map-world-1>

Remote sensing analysis can be triggered right after a sudden-onset hazard, and preliminary findings can be made available almost immediately following a shock. A powerful tool for resource mobilization and to get DIEM analysis in front of partners rapidly is a StoryMap. These live products include interactive maps and analysis to help stakeholders target response, and immediately understand the needs of the communities.

In response to demand from the field, DIEM developed the **Events Visualization in Emergencies (EVE)** tool to enhance the understanding of flood events and their impacts on various land cover types, focusing on agricultural areas. Thanks to biweekly EVE updates, DIEM now produces a series of short products focused on flooding and other hazards in West Africa and the Sudan, among other countries. These products are especially vital to ensuring that partners and stakeholders understand where and when to target response.

DIEM also contributes to **DIEM-Risk**, which strengthens risk assessments to inform disaster risk reduction and anticipatory action with products such as flood exposure maps and risk scores; and **DIEM-Research**, which

provides action-oriented research to improve support to agricultural livelihoods in fragile and risk-prone contexts. A small programme providing research grants has been established to foster the use of DIEM data by several universities across Africa.

DIEM is a unique tool that enables FAO and other partners to direct programming to households based on their needs. Myriad examples now point to DIEM's added value. Following the October 2023 earthquakes in Afghanistan, DIEM conducted a rapid assessment utilizing satellite images – which was followed up by ground-truthing field surveys – and published a **StoryMap** within 72 hours of the shock. The StoryMap directed stakeholders to the most impacted areas and provided response recommendations. The FAO Representation in Afghanistan was able to immediately utilize the StoryMap for resource allocation and to direct response based on the initial recommendations. The identified needs and related response demonstrate how DIEM is able to illustrate pathways from evidence to programme through products like StoryMaps. DIEM data continue to prove their importance and use to programming around the world.

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Contact

Office of Emergencies and Resilience
data-in-emergencies@fao.org | data-in-emergencies.fao.org

Food and Agriculture Organization of the United Nations
 Rome, Italy